Cardiorespiratory OSCE Prep

Cases E & F · Physiotherapy · Target: Very Good (≥70%)

CASE E

Mr. Liam O'Connor, 19M — Cystic Fibrosis (Pulmonary Tune-up)

FEV_1 decline · Increased purulent sputum volume & viscosity · Minor haemoptysis · Home PEP + exercise therapy · Poor adherence (university)

1 SUBJECTIVE ASSESSMENT

CONSENT / OPENING STATEMENT

> *"Hi Liam, my name is [Name], I'm the physiotherapist looking after you today. I'd like to ask you some questions about your breathing and chest, then look at your current airway clearance routine. Everything stays within your care team. Is that okay, and do you have any questions before we start?"*

KEY QUESTIONS TO ASK

"Can you describe your sputum — how much, what colour, and has it changed recently?"
I ask this to quantify secretion load and gauge infection severity. Increased purulent sputum (green/yellow) indicates active Pseudomonas exacerbation; volume guides intensity of airway clearance.

"Tell me about the blood — is it streaked or a larger volume? Does it happen after coughing hard?"
I ask this to risk-stratify haemoptysis. Minor streaking differs critically from frank haemoptysis (>240 mL/24 h), which requires immediate medical escalation and cessation of all airway clearance.

"Walk me through your current PEP routine — pressure setting, frequency, how many cycles, how long?"
I ask this to identify technique errors or adherence gaps. Incorrect OPEP pressure or shortened sessions significantly reduce secretion mobilisation efficacy and may explain ongoing symptoms.

"How is university affecting your therapy — which sessions do you most often skip and why?"
I ask this to identify specific adherence barriers so I can tailor a realistic, schedule-compatible plan. Non-adherence is the leading cause of FEV_1 decline in young CF adults.

"On a 0–10 scale, how breathless do you get during exercise and at what point do you stop?"
I ask this to establish a Borg baseline and understand whether exercise limitation is ventilatory, musculoskeletal, or motivational — guiding safe exercise prescription.

"Are you on nebulised DNase or hypertonic saline — and do you take them before or after physio?"
I ask this because mucolytics must precede airway clearance by at least 30 minutes to maximise sputum mobilisation. Incorrect sequencing significantly reduces treatment effectiveness.

RED FLAGS

Red Flag	Significance	Action if Present
Frank Haemoptysis (>240 mL/24 h)	Bronchial artery erosion; life-threatening airway haemorrhage risk	Cease all airway clearance; urgent medical escalation; lateral decubitus (bleeding side down)
SpO_2 <88% at rest or with activity	Severe hypoxaemia; risk of cor pulmonale and acute deterioration	Halt exercise; apply supplemental O_2; notify medical team; do not proceed without review
Pneumothorax signs (sudden pleuritic pain, absent unilateral breath sounds, tracheal deviation)	CF patients have elevated pneumothorax risk; tension pneumothorax is fatal without urgent intervention	Emergency call; cease all manual techniques and PEP immediately
New haemodynamic instability (HR >130 at rest, BP drop)	Suggests septic exacerbation or pulmonary embolism	Medical review before any physiotherapy; document and monitor vitals
Severe FEV_1 decline (<30% predicted) or Borg ≥7 at minimal exertion	Near end-stage disease; standard exercise and PEP must be heavily modified	Reduce intensity; consider NIV support; engage palliative/transplant team

PROBLEM LIST

PHYSICAL IMPAIRMENTS

- Impaired mucociliary clearance → retained secretions
- Airflow obstruction (↓ FEV_1)
- Increased mucus viscosity (CFTR dysfunction)
- Reduced exercise capacity / deconditioning

FUNCTIONAL ACTIVITY LIMITATIONS

- Unable to complete home PEP without exhaustion
- Exercise tolerance reduced (dyspnoea-limited)

- Minor haemoptysis (fragile inflamed bronchial mucosa)

- Difficulty maintaining therapy around university schedule
- Risk of further FEV_1 decline if adherence not addressed

2 PATHOPHYSIOLOGY

Core defect: Mutations in the CFTR gene produce dysfunctional or absent chloride channels on airway epithelium. This leads to dehydrated, hyperviscous airway surface liquid — mucus that cannot be cleared by normal mucociliary action.

Infection-inflammation cycle: Retained mucus plugs obstruct small and medium airways, creating a nidus for chronic infection — predominantly *Pseudomonas aeruginosa*. Neutrophil-dominated inflammation releases elastase and proteases that destroy airway walls, driving bronchiectasis, air trapping, and progressive obstructive lung disease.

FEV_1 decline mechanism: Each exacerbation accelerates bronchiectatic remodelling, reducing elastic recoil and increasing resistance. The cycle of infection → inflammation → structural damage compounds over time.

Exercise limitation: Increased dead space ventilation, dynamic hyperinflation, and impaired gas exchange create ventilatory limitation well below peak cardiac capacity. Deconditioning from reduced activity further reduces peak VO_2.

Haemoptysis mechanism: Chronic inflammation drives neovascularisation of bronchial arteries, which are fragile and prone to rupture with forceful coughing during exacerbations. Minor haemoptysis indicates mucosal inflammation, not necessarily structural erosion.

Why Liam fits: Purulent sputum (active bacterial infection), exhaustion with PEP (ventilatory limitation from ↓ FEV_1 and dynamic hyperinflation), and haemoptysis (inflamed neovascularised mucosa) all map directly onto this cycle of infection → inflammation → structural decline.

3 DIFFERENTIAL DIAGNOSES

Differential 1: Allergic Bronchopulmonary Aspergillosis (ABPA)

- Common CF complication (prevalence 2–15%)
- Presents with worsening airflow obstruction, increased mucus, and constitutional symptoms
- Streaky haemoptysis and elevated total IgE may occur
- Thick brown mucus plugs are a classic feature

I favour CF exacerbation over ABPA because... Liam's sputum is purulent (green/yellow) rather than the brown plug-like mucus typical of ABPA. ABPA presents with eosinophilia, elevated total and Aspergillus-specific IgE — none flagged here. His clinical picture is consistent with bacterial exacerbation, the most common reason for CF tune-up admissions.

Differential 2: Non-CF Bronchiectasis (e.g., post-infective)

- Shares features: chronic productive cough, purulent sputum, recurrent chest infections
- Similar airflow obstruction pattern and exercise intolerance
- Haemoptysis can occur in bronchiectasis of any aetiology

I favour CF over non-CF bronchiectasis because... Liam has a confirmed CF diagnosis with established CFTR dysfunction and a childhood history of disease. His entire management history — OPEP therapy, mucolytics, recurrent tune-up admissions — is CF-specific. Non-CF bronchiectasis does not account for his CFTR genotype, the viscosity profile of his sputum, or his documented home PEP regimen.

4 OBJECTIVE ASSESSMENT & SPECIAL TESTS

KEY OBSERVATIONS

RR: target ≤25/min at rest | **SpO_2: ≥93% on air (exacerbation baseline)**

Accessory muscle use → ↑ WOB | **Sputum: volume, colour, consistency**

Chest shape: barrel / hyperinflation

AUSCULTATION FINDINGS EXPECTED

- Widespread coarse crackles bilaterally — retained secretions in bronchiectatic airways
- Expiratory wheeze — dynamic airway collapse and obstruction
- Reduced air entry in lower zones — mucus plugging and air trapping
- Post-clearance: crackle reduction = effective treatment (use as immediate outcome feedback)

SPECIAL TESTS

Test 1: Spirometry — FEV_1 / FVC (Pre & Post Bronchodilator)

WHAT IT MEASURES

Degree of airflow obstruction; tracks decline from personal best; assesses bronchodilator reversibility

HOW TO PERFORM

Patient seated, noseclip applied. Maximal inspiration then rapid, forceful, sustained expiration for ≥6 seconds. Repeat 3 acceptable manoeuvres; take the best of 3. Repeat 15 minutes after salbutamol 400 mcg.

POSITIVE FINDING

FEV_1/ FVC <0.70 (obstructive pattern); FEV_1 <70% predicted; <12% improvement post-bronchodilator = fixed obstruction typical of CF.

WHY I CHOSE THIS

"I select spirometry because FEV_1 is the primary marker of CF disease severity and the gold standard for monitoring exacerbation response. It directly quantifies the obstruction driving Liam's exercise limitation and guides treatment intensity."

Test 2: Six-Minute Walk Test (6MWT) or Modified Shuttle Walk Test

WHAT IT MEASURES

Functional exercise capacity; oxygen desaturation on exertion; baseline for exercise rehabilitation

HOW TO PERFORM

10- metre flat corridor (6MWT) or pre-recorded beep levels (MSWT). Monitor SpO_2, HR, and Borg (0–10) at each minute. Record distance achieved. Stop if SpO_2 <88%, Borg ≥8, or patient requests. Repeat post-treatment for comparison.

POSITIVE FINDING

<82% predicted 6MWT distance = impaired functional capacity. SpO_2 drop >4% with exertion = exercise-induced hypoxaemia requiring O_2 review.

WHY I CHOSE THIS

"I select this because Liam reports exercise exhaustion — I need an objective baseline to safely prescribe exercise and monitor rehabilitation progress. It also identifies whether supplemental O_2 is needed during exertion."

Test 3: OPEP Technique Assessment (Aerobika / Acapella)

WHAT IT MEASURES

Accuracy of current home airway clearance technique; identifies technique errors contributing to inadequate secretion mobilisation

HOW TO PERFORM

Ask Liam to demonstrate his OPEP routine. Observe: upright posture, tidal breathing through device (not forced), resistance setting (10–20 cmH_2O), number of cycles (8–10 breaths per set), huff coughing after each set, and session duration (20–30 min). Auscultate before and after.

POSITIVE FINDING

Incorrect pressure, forced breathing pattern, or shortened sessions = technique error. Reduction in crackles post-session = effective clearance response.

WHY I CHOSE THIS

"I select technique assessment because Liam's reported exhaustion with PEP may reflect incorrect resistance settings requiring excessive effort, or incomplete sessions. Identifying technique errors is essential before modifying his plan."

OUTCOME MEASURES

- **FEV_1 % predicted** — primary CF outcome; target return to personal best
- **LCQ (Leicester Cough Questionnaire)** — patient-reported QoL impact of cough and secretions
- **Borg Scale (0–10)** — dyspnoea and fatigue during exercise and airway clearance sessions

5 TREATMENT PLAN

INITIAL PHASE — WEEKS 1–2 (INPATIENT TUNE-UP)

AIRWAY CLEARANCE: OSCILLATING PEP (AEROBIKA / ACAPELLA)

How to do: Seated slightly forward-leaning. Tidal breathing through device at 10–20 cmH_2O resistance for 8–10 breaths → 2–3 huffs (FET) → 30-sec rest. Repeat 3–4 cycles. Total 20–30 min. Perform after nebulised DNase/hypertonic saline (minimum 30 min post-nebs).

↓ REGRESSION	STANDARD	↑ PROGRESSION
5–6 breaths per cycle; lower resistance setting; seated in recliner if fatigued; 1 session/day during severe exacerbation	2 sessions/day (morning + evening); 8–10 breaths × 4 cycles; huff cough × 2–3 after each set	3 sessions/day; add ACBT as adjunct; incorporate postural drainage for lower lobe clearance

BREATHING EXERCISE: ACTIVE CYCLE OF BREATHING TECHNIQUE (ACBT)

How to do: 3-phase cycle — (1) Breathing control: gentle diaphragmatic breathing × 4–6 breaths; (2) Thoracic expansion exercises: 3–4 slow deep breaths with 3-sec inspiratory hold; (3) FET/huff: 1–2 medium-volume huffs to shift secretions centrally. Repeat 3–4 cycles; finish with effective cough.

↓ REGRESSION	STANDARD	↑ PROGRESSION
Breathing control + gentle huff only; avoid deep expansion if SpO_2 <90%	Full ACBT cycle 2× per day; combined with OPEP as primary modality	Add postural drainage positions; introduce autogenic drainage; independent self-management

EXERCISE / MOBILITY

Starting level: 10–15 min supervised walking on ward at Borg ≤4; stationary bike or arm ergometry at 50–60% peak HR if SpO_2 stable.

↓ REGRESSION	STANDARD	↑ PROGRESSION
Seated arm raises, leg marching; 5–10 min with rest between; if SpO_2 <90% — rest only	20–30 min aerobic activity; 3–5× per week; Borg 4–6 target zone	Interval training; resistance exercise; unsupervised community programme; sport participation

PATIENT EDUCATION — 4 KEY MESSAGES

1. **Sequencing matters:** Always use mucolytics (DNase, hypertonic saline) 30–60 min before airway clearance — this dramatically improves sputum mobilisation and makes every session more effective.
2. **Adherence strategy:** Two sessions daily is the evidence-based minimum; we will map sessions to your lecture timetable — a morning and evening 20-minute routine is realistic even around a full university day.
3. **Exercise is treatment, not extra:** Aerobic exercise augments mucociliary clearance and slows FEV_1 decline — it is a core part of CF management, not optional.
4. **Haemoptysis protocol:** Minor streaking during coughing is common in CF. If blood volume increases significantly, you feel unwell, or it occurs outside coughing, stop all airway clearance and contact the team immediately.

> **Progression Criteria (to Phase 2):**
> SpO_2 ≥93% at rest · Borg ≤4 during 15-min walk · Sputum volume and purulence reducing · FEV_1 trending toward personal best · Self-managing OPEP technique correctly and independently

LONG-TERM — WEEKS 4–12 (OUTPATIENT / SELF-MANAGEMENT)

- **CF Pulmonary Rehabilitation:** Supervised aerobic + resistance programme 2–3× per week; target ≥150 min moderate-intensity exercise per week
- **Adherence tools:** CF diary app (e.g., myCF) to log sessions; phone reminders mapped to university timetable; involve CF nurse specialist
- **Airway clearance independence:** Self-managing OPEP + ACBT; selects appropriate technique based on daily secretion volume
- **Annual review:** Spirometry tracked against personal best; written exacerbation action plan in place
- **Discharge criteria:** FEV_1 ≥85% personal best · SpO_2 ≥93% at rest · Independent airway clearance ×2/day · Functional exercise capacity at pre-admission level · University schedule plan documented

Examiner Demonstration — I can show in the room:

1. **Oscillating PEP technique:**

   Demonstrate correct posture, tidal breathing through device, resistance check, transition to huff cough, and repeat cycle — with verbal cues throughout

2. **Huff coughing / FET:**

   Demonstrate open-glottis medium-volume huff vs. high-volume huff, and explain when each is used to mobilise vs. expectorate secretions centrally

CASE F

Mr. Robert Miller, 64M — Day 2 Post-CABG (Median Sternotomy)

30 pack-year smoking · Mild baseline COPD · Sternal pain / "clicking" · Chest drain + telemetry · Shallow apical breathing · Weak guarded cough · Goal: prevent PPCs & early mobilisation

1 SUBJECTIVE ASSESSMENT

CONSENT / OPENING STATEMENT

"Good morning Mr. Miller, my name is [Name], I'm the physiotherapist on the cardiac surgery ward. I'm here to help you with your breathing and get you moving safely after your operation. I know you're very sore — that is completely expected — and everything we do today will be at your pace and comfort level. Is that alright? Do you have any concerns before we start?"

KEY QUESTIONS TO ASK

"Can you describe your pain — where exactly, score out of 10, and when did you last take your analgesia?"
I ask this because pain management is a prerequisite for effective post-sternotomy physiotherapy. Uncontrolled pain directly causes shallow breathing, guarded cough, and increased PPC risk. I need to time therapy 30–45 min after analgesia peak.

"When you say the sternum is 'clicking' — can you tell me more? Does it click with every breath, only with movement, or only when you cough?"

I ask this to assess sternal stability. Clicking with every breath or at rest suggests possible dehiscence — a red flag requiring surgical review before any physiotherapy. Clicking only with coughing may be gas or tissue movement and is more likely benign early post-operatively.

"Can you take a deep breath, or does pain stop you before you get there?"
I ask this to establish current ventilatory capacity and identify pain-splinting behaviour. Inadequate tidal volume drives atelectasis — the primary mechanism of post-operative pulmonary complications.

"Before the operation, did you have a regular morning cough, bring up sputum, or get breathless with everyday activities?"
I ask this to establish pre-operative COPD baseline. Productive cough pre-operatively indicates greater secretion retention risk post-surgery, requiring more active airway clearance strategies.

"How many cigarettes were you smoking before admission — and do you still want to smoke now?"
I ask this to assess current smoking status and readiness to quit. Post-CABG is a high-leverage cessation opportunity; continued smoking impairs wound healing, ciliary function, and graft patency, worsening PPC risk.

"What are you most worried about with doing exercises today — is it the pain, the tubes, or something else?"
I ask this to identify the specific psychological barrier to mobilisation. Targeted reassurance (sternal dehiscence fear vs. pain vs. drain displacement) is more effective than generalised encouragement at improving patient engagement.

RED FLAGS

Red Flag	Significance	Action if Present
Sternal Dehiscence (visible gap, paradoxical movement, clicking with every breath, wound instability)	Loss of sternal integrity; risk of mediastinitis, cardiac tamponade, and catastrophic haemorrhage	Cease all physiotherapy immediately; keep patient still; urgent surgical team review; do not apply sternal support cough
New Arrhythmia / ECG Changes on Telemetry (AF, VT, ST changes)	Post-CABG arrhythmias (AF in 25–40%); may indicate graft ischaemia or electrolyte disturbance	Halt exercise; confirm rhythm on monitor; notify medical team; do not mobilise until reviewed
SpO_2 <90% or RR >30/min at rest	Significant hypoxaemia suggesting extensive atelectasis, pulmonary embolism, or pulmonary oedema	Escalate supplemental O_2; urgent medical review; modify or cease physiotherapy pending review

Red Flag	Significance	Action if Present
Chest Drain: sudden increase in output (bright red blood) or abrupt cessation with deteriorating obs	Haemothorax (high output) or cardiac tamponade (cessation) — both surgical emergencies	Do not mobilise; immediate surgical team notification
DVT / PE signs (unilateral leg swelling, sudden pleuritic chest pain, SpO_2 drop, tachycardia)	Post-surgical hypercoagulable state; PE is a leading cause of post-operative mortality	Halt all mobility; emergency medical assessment; anticoagulation review urgently

PROBLEM LIST

PHYSICAL IMPAIRMENTS

- Post-sternotomy pain → splinting → basal atelectasis
- Reduced chest wall compliance post-surgery
- Shallow apical breathing (impaired diaphragm recruitment)
- Weak, guarded, ineffective cough
- Mild baseline COPD (pre-existing airflow obstruction)
- Sternal instability / "clicking" — requires characterisation

FUNCTIONAL ACTIVITY LIMITATIONS

- Reluctant to mobilise, breathe deeply, or cough
- Unable to clear secretions effectively
- Bed-bound → DVT risk, deconditioning
- Fear-avoidance behaviour limiting recovery participation
- Mobility limited by chest drain and telemetry attachments

2 PATHOPHYSIOLOGY

Surgical mechanism of PPC: Median sternotomy divides the sternum and retracts the chest wall, causing direct thoracic cage trauma. Combined with general anaesthesia, cardiopulmonary bypass, supine positioning, and post-operative pain, this produces a predictable pattern of reduced tidal volume, loss of functional residual capacity (FRC), and progressive alveolar collapse — predominantly in dependent basal zones.

Atelectasis cascade: Surfactant dysfunction from bypass, diaphragmatic splinting from pain, and impaired cough reflex allow distal airways to collapse. Gravitational forces compound basal zone collapse in supine positioning. The result is ventilation/perfusion (V/Q) mismatch, hypoxaemia, and a fertile environment for post-operative pneumonia.

COPD additive risk: Robert's mild COPD means he has reduced mucociliary clearance, increased mucus production, and already-compromised small airway function at baseline.

General anaesthesia and surgical stress further impair these defences, placing him at significantly greater risk of PPCs than a non-COPD patient.

Pain–breathing feedback loop: Pain with inspiration → splinting → shallow breathing → atelectasis → hypoxaemia → anxiety → more splinting. Breaking this cycle with adequate analgesia and supported deep breathing is the primary physiotherapy goal in the early post-operative period.

Haemodynamic context: Post-CABG patients are at risk for arrhythmias (particularly AF in the first 48–72 hours), fluid overload (pulmonary oedema), and early graft complications. Physiotherapy must be timed and dosed to avoid excessive myocardial demand.

Why Robert fits: His shallow apical breathing, guarded cough, reluctance to move, and reduced basal breath sounds all map directly to the atelectasis-pain-splinting cycle. His 30 pack-year history and COPD compound the baseline risk for secretion retention and PPC development.

3 DIFFERENTIAL DIAGNOSES

Differential 1: Post-operative Pneumonia

- Common PPC following cardiac surgery (incidence 2–22%)
- Presents with productive cough, fever, CXR consolidation, raised WCC
- May develop from retained secretions, aspiration during intubation, or haematogenous spread
- Baseline COPD significantly increases susceptibility

I favour basal atelectasis over pneumonia because... Robert is only 2 days post-op — the typical time frame for atelectasis rather than established pneumonia, which usually manifests with fever, purulent sputum, and CXR consolidation at 2–5+ days. His reduced chest expansion and shallow breathing are mechanical, consistent with pain-splinting rather than infective consolidation. CXR should be reviewed to differentiate, but the clinical signs point to atelectasis as the primary pathology at this stage.

Differential 2: Pulmonary Embolism (PE)

- Post-surgical hypercoagulable state + immobility = significantly elevated DVT/PE risk
- Presents with pleuritic chest pain, tachycardia, dyspnoea, and sudden SpO_2 drop
- Sternal pain could mask pleuritic PE pain in this patient
- Sudden-onset respiratory deterioration differentiates PE from gradual atelectasis

I favour atelectasis over PE because... Robert's symptoms are bilateral, gradual in onset, and consistent with diffuse basal collapse from pain-splinting — not the sudden-onset,

unilateral, haemodynamically significant presentation typical of PE. He is on continuous telemetry with no sudden deterioration documented. PE remains an active monitoring priority — any sudden SpO_2 drop, new tachycardia, or unilateral leg swelling warrants urgent review before any physiotherapy proceeds.

4 OBJECTIVE ASSESSMENT & SPECIAL TESTS

KEY OBSERVATIONS

RR: expected 18–24/min (splinting) | **SpO_2: target ≥94% on O_2** | **HR: telemetry — watch for arrhythmia** | **Sternal wound: dressing intact, no separation** | **Chest drain: character and volume of output** | **Accessory muscle use → ↑ WOB**

AUSCULTATION FINDINGS EXPECTED

- Diminished or absent breath sounds bilaterally at bases — basal atelectasis
- Fine inspiratory crackles — alveoli opening on deep inspiration (positive sign of response)
- Coarse crackles if secretion retention present — more likely in COPD patient
- Upper zone breath sounds preserved — consistent with apical breathing pattern
- Post-deep breathing exercise: improved basal air entry and reduced crackles = treatment effectiveness

SPECIAL TESTS

Test 1: Chest Expansion Measurement

WHAT IT MEASURES

Bilateral chest wall excursion; identifies basal restriction due to pain and atelectasis; quantifies improvement with treatment

HOW TO PERFORM

Patient seated or semi-reclined. Tape measure at 4th intercostal space (mid-chest) and at xiphisternum (lower chest). Record circumference at full expiration and full inspiration; calculate difference. Normal ≥5 cm. Assess symmetry with bilateral hand placement — unequal expansion suggests unilateral pathology.

POSITIVE FINDING

<3– 4 cm expansion at bases = significant restriction from pain-splinting or atelectasis. Asymmetrical expansion = unilateral pathology (effusion, pneumothorax).

WHY I CHOSE THIS

"I select chest expansion measurement because it gives an objective, repeatable measure of breathing restriction from sternal pain and splinting, and quantifies improvement after breathing

exercises — guiding which lung zones to target."

Test 2: Incentive Spirometry Assessment

WHAT IT MEASURES

Sustained maximal inspiratory effort; degree of atelectasis and inspiratory muscle weakness; visual biofeedback tool

HOW TO PERFORM

Patient in high Fowler's (60–80°). Apply sternal support pillow first. Instruct slow maximal inhalation through IS mouthpiece to raise and hold the piston for 3–5 seconds. Record volume achieved (mL). Target ≥70% predicted inspiratory volume. Prescribe 10 repetitions each hour while awake.

POSITIVE FINDING

<50% predicted inspiratory volume = significant restriction. Inability to sustain inhalation for 3 seconds = impaired inspiratory muscle endurance.

WHY I CHOSE THIS

"I select incentive spirometry because it directly addresses the atelectasis mechanism by encouraging sustained alveolar inflation. It gives me a visual volume baseline to track, and Robert can use it independently between physio sessions."

Test 3: Sternal Stability Assessment

WHAT IT MEASURES

Integrity of sternal wiring post-sternotomy; determines safety to proceed with coughing, upper limb exercise, and progressive mobilisation

HOW TO PERFORM

Observe wound for visible gap, haematoma, or discharge. Palpate gently around sternal edges for crepitus, mobility, or asymmetry. Ask patient to perform a gentle cough — observe sternal movement. Characterise clicking: location, timing relative to breathing and movement. Cross-reference surgical notes and nursing observations.

POSITIVE FINDING (CONCERNING)

Palpable sternal movement, visible separation, or clicking with every breath at rest = possible dehiscence → halt physiotherapy immediately, notify surgeon.

WHY I CHOSE THIS

"I select sternal stability assessment because Robert's reported 'clicking' must be characterised before I apply sternal support coughing or any exercise that loads the sternum. Confirming sternal integrity is a non-negotiable safety prerequisite for all subsequent interventions."

OUTCOME MEASURES

- **Incentive spirometry volume (mL)** — direct inspiratory capacity measure; track daily improvement toward predicted volume
- **6-Minute Walk Test** — functional mobility baseline before discharge; compare to pre-operative level
- **Borg Scale (0–10)** — dyspnoea and pain during activity; guides progression decisions
- **CXR** — medical outcome measure; track atelectasis resolution (report findings back to medical team)

5 TREATMENT PLAN

INITIAL PHASE — DAYS 1–7 (ICU STEP-DOWN / CARDIAC WARD)

> **Pre-treatment safety check:** Confirm analgesia administered 30–45 min prior · Review telemetry (no new arrhythmia) · Check chest drain patency and output · Confirm sternal stability · Only then commence physiotherapy.

BREATHING EXERCISE: DIAPHRAGMATIC BREATHING + BASAL EXPANSION

How to do: Patient in high Fowler's (60–80°). Apply sternal support pillow. Place hands on lower lateral ribs. Instruct slow nasal inhalation, expanding laterally against hands for 3–4 sec, 2-sec hold, slow mouth expiration 4–6 sec. Focus on basal/lateral expansion — not apical. 10 repetitions, 4–6 sessions per day. Progress to incentive spirometry 10 reps each hour awake.

↓ REGRESSION	STANDARD	↑ PROGRESSION
Supine; therapist hand on abdomen for proprioceptive feedback; 5 breaths then rest; reduce depth if SpO_2 drops	High Fowler's; 10 reps × 4–6 sessions/day; combined with IS ×10 reps each waking hour	Sitting unsupported; upper limb elevation with inspiration; walking with coordinated breathing; increasing IS volume targets

AIRWAY CLEARANCE: STERNAL-SUPPORTED COUGH (PILLOW SPLINTING)

How to do: Only after confirming sternal stability AND adequate analgesia. Patient holds firm pillow firmly against sternum with both arms crossed. Takes a deep breath in, then produces a controlled huff cough (not an explosive cough). Begin with huff technique; escalate to 2-stage cough as pain reduces. Do not allow unsupported vigorous coughing in the early post-operative period.

↓ REGRESSION	STANDARD	↑ PROGRESSION
Huff only (no cough); therapist-supported; minimal force; 2–3 attempts then rest if SpO_2 drops or pain ≥5/10	Self-supported pillow cough; patient-directed timing; following each breathing session	Unsupported cough as sternal healing progresses (4–6 weeks); ACBT if COPD-related sputum is a persistent issue

EXERCISE / MOBILITY

Starting level (Day 2 post-CABG): Sit out of bed → stand with assistance → 2-minute supervised walk in room. Ankle pumps and lower limb circulation exercises in bed between sessions.

↓ REGRESSION	STANDARD	↑ PROGRESSION
Sit out of bed only (legs dependent); active limb exercises in bed; if haemodynamically unstable — bed exercises only	Supervised ward walks ×2–3/day, increasing distance each session; sit-to-stand practice; stair practice before discharge	Independent ambulation; stairs; home walking programme; outpatient cardiac rehabilitation

PATIENT EDUCATION — 4 KEY MESSAGES

1. **Sternal precautions (8–12 weeks):** No pushing or pulling with arms — no steering wheel pushing, no lifting from the floor. Always hug the pillow when coughing. If the sternum feels unstable or clicking worsens, stop and call the nurse immediately.
2. **Deep breathing is medicine:** Shallow breathing is the main cause of lung complications after heart surgery. Ten deep breaths with the incentive spirometer every hour while awake is one of the most important things Robert can do to prevent pneumonia.
3. **Smoking cessation:** This is the single most impactful change Robert can make post-CABG — cessation reduces re-infarction risk, improves wound healing, restores ciliary function, and prolongs graft patency. A formal cessation referral will be arranged.
4. **Cardiac rehabilitation:** A structured 6–8 week outpatient cardiac rehab programme significantly reduces mortality and improves quality of life after CABG — Robert will be referred before discharge.

> **Progression Criteria (to Phase 2):**
>
> SpO_2 ≥94% on ≤2L O_2 or room air · RR ≤20/min at rest · Pain ≤4/10 at rest · Walking corridor independently · Chest drain removed · Sternal stability confirmed · Effective supported cough independently performed

LONG-TERM — WEEKS 2–12 (OUTPATIENT / CARDIAC REHAB)

- **Cardiac Rehabilitation (Phase II & III):** Supervised aerobic exercise (walking, cycling) at 60–80% HR reserve; telemetry initially; 6–8 week structured programme with individualized progression
- **COPD management:** Spirometry review at 6–8 weeks post-discharge; optimise inhaler therapy; smoking cessation programme ongoing with follow-up
- **Home exercise programme:** Daily walking target starting at 10–15 min, building to ≥30 min; sternal precaution education reinforced until 12 weeks post-surgery

- **Discharge criteria:** Ambulating ≥200 m independently without desaturation · Pain controlled on oral analgesia only · Sternal wound healed and stable · Independently performing breathing exercises · Cardiac rehab referral confirmed · Smoking cessation support engaged

Examiner Demonstration — I can show in the room:

1. **Pillow sternal support cough:**

   Demonstrate correct pillow placement against the sternum, arm-crossing position, instruction for controlled huff cough vs. explosive cough, and how to teach the patient to self-manage this technique independently

2. **Basal expansion breathing with proprioceptive cues:**

   Demonstrate hand placement on lower lateral ribs, instruction for lateral costal breathing vs. apical breathing, and coordination with incentive spirometry volume targets as immediate feedback